First Order ODEs

The **general form** of a first order ODE is

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$$

for some given function f.

1 Linear

The **general form** of a first order linear ODE is

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t)$$

for some given functions p and q.

1.1 Method of Integrating Factors

For ODEs of the form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t)$$

the **general solution** is

$$y = \frac{1}{\mu(t)} \int \mu(t) \cdot q(t) \, \mathrm{d}t$$

where the **integrating factor** is any $\mu(t)$ that satisfies

$$\ln |\mu(t)| = \int p(t) \, \mathrm{d}t$$

Remark. Indefinite integrals yield a family of functions differing by a constant. Each indefinite integral results in an independent constant, which represents a degree of freedom. However, the constant from $\mu(t)$ cancels out in the final solution for y.

1.2 Variation of Parameters

For ODEs of the form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t)$$

let $h(t)$ be a function such that

$$h(t) = \int p(t) \, \mathrm{d}t$$

Then the **general solution** is given by

$$y = e^{-h(t)} \int q(t) \cdot e^{h(t)} \, \mathrm{d}t$$

1.3

For ODEs of the form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t)$$

the **general solution** is given by

$$y = v \int \frac{q(t)}{v} \, \mathrm{d}t$$

where v satisfies

$$\frac{\mathrm{d}v}{\mathrm{d}t} + p(t) \cdot v = 0$$

Proof.

We want to solve eq. (1). Suppose v satisfies eq. (2), which is separable and can be solved easily.

$$y' + p(t) \cdot y = q(t) \tag{1}$$

$$v' + p(t) \cdot v = 0 \tag{2}$$

Take the derivative of y/v:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{y}{v}\right) &= \frac{y'v - v'y}{v^2} \\
&= \frac{y'v + (p(t) \cdot v)y}{v^2} && \text{Plug in eq. (2).} \\
&= \frac{q(t)}{v} && \text{Plug in eq. (1).}
\end{aligned}$$

Thus we have

$$\frac{y}{v} = \int \frac{q(t)}{v} \, \mathrm{d}t$$

Since v can be solved from eq. (2), we obtain the solution for y. □

2 Non-Linear

2.1 Separable Equations

A first order ODE is **separable** if it can be written in the form

$$M(t) + N(y) \cdot \frac{\mathrm{d}y}{\mathrm{d}t} = 0$$

for some functions M and N. Then the solution satisfies

$$\int N(y) \, \mathrm{d}y = -\int M(t) \, \mathrm{d}t$$

2.2 Transformation Methods

Some complicated non-linear ODEs can be transformed into ODEs that are solvable by suitable substitutions.

2.2.1 Bernoulli Equations

Bernoulli equations can be transformed into linear ODEs by the substitution $v = y^{1-r}$.

Definition 2.1 (Bernoulli Equations)**.**

A first order ODE is called a **Bernoulli equation** if it can be written in the form

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t) \cdot y^r \tag{3}$$

for some given functions p, q and a real number $r \in \mathbb{R} \setminus \{0, 1\}$.

The **general solution** of a Bernoulli equation (3) is given by

$$y^{1-r} = \frac{1-r}{\mu(t)} \int \mu(t) \cdot q(t) \, \mathrm{d}t$$

where $\mu(t)$ satisfies

$$\ln|\mu(t)| = (1-r) \int p(t) \, \mathrm{d}t$$

Note: If $r > 0$, then $y = 0$ is also a solution, which is not captured by the above formula.

Proof.

Substitute $v = y^{1-r}$, and then solve with the method of integrating factors.

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y = q(t) \cdot y^r \quad &\implies \quad y^{-r}\frac{\mathrm{d}y}{\mathrm{d}t} + p(t) \cdot y^{1-r} = q(t) \quad \text{for} \quad y \neq 0 \\
&\implies \quad \frac{\mathrm{d}}{\mathrm{d}t}\left(y^{1-r}\right) + (1-r)p(t) \cdot y^{1-r} = (1-r)q(t)
\end{aligned}$$

□

2.2.2 Homogeneous Equations

Homogeneous equations can be transformed into separable equations.

If a first-order ODE can be expressed in the following form, then it is said to be **homogeneous**:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = F(y/t)$$

for some given function F.

Then it can be transformed into a separable equation using $v = y/t$:

$$\frac{1}{F(v) - v}\,\mathrm{d}v = \frac{1}{t}\,\mathrm{d}t$$

Remark. The homogeneous equations considered here have nothing to do with the homogeneous LCCDEs.

2.3 Exact Equations

Definition 2.2 (Exact Equations)**.**

A first order ODE $M(t, y)\,\mathrm{d}t + N(t, y)\,\mathrm{d}y = 0$ is said to be **exact** if there exists a function $\Psi(t, y)$ such that

$$\frac{\partial \Psi(t, y)}{\partial t} = M(t, y) \quad \text{and} \quad \frac{\partial \Psi(t, y)}{\partial y} = N(t, y)$$

Then the general solution is given implicitly as

$$\Psi(t, y) = C \qquad \text{for some constant } C \in \mathbb{R}$$

Theorem 2.1.

Let R be a simply connected region in the ty-plane. If $M(t, y)$, $N(t, y)$, $M_y(t, y)$, and $N_t(t, y)$ are continuous in R, then

$$M(t, y) + N(t, y)\frac{\mathrm{d}y}{\mathrm{d}t} = 0 \text{ is exact in } R \qquad \iff \qquad M_y(t, y) = N_t(t, y) \quad \text{for all } (t, y) \in R$$

where $M_y(t, y) := \partial M(t, y)/\partial y$ and $N_t(t, y) := \partial N(t, y)/\partial t$.

To solve the first order ODE $M(t, y)\,\mathrm{d}t + N(t, y)\,\mathrm{d}y = 0$, we can follow these steps:

1. Verify that the equation is exact by checking if $M_y(t, y) = N_t(t, y)$.
2. Integrate $M(t, y)$ with respect to t to find $\Psi(t, y)$:

$$\Psi(t, y) = \int M(t, y)\,\mathrm{d}t + h(y)$$

   where $h(y)$ is an arbitrary function of y.

3. Differentiate $\Psi(t, y)$ with respect to y and set it equal to $N(t, y)$:

$$\frac{\partial \Psi(t, y)}{\partial y} = N(t, y)$$

This will allow us to solve for $h(y)$.

4. Substitute $h(y)$ back into $\Psi(t, y)$ to obtain the implicit solution:

$$\Psi(t, y) = C$$

for some constant C.

2.4 Exact Equations and Integrating Factors

It is sometimes possible to convert an ODE that is not exact into an exact equation with a suitable integrating factor. Suppose the following first order ODE is exact:

$$\mu M(t, y) + \mu N(t, y)\frac{\mathrm{d}y}{\mathrm{d}t} = 0$$

Then the integrating factor μ must satisfy

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial t}$$

On one hand, assuming that $\mu = \mu(t)$, we have

$$\frac{\mathrm{d}\mu}{\mathrm{d}t} = \frac{M_y - N_t}{N} \cdot \mu$$

If $(M_y - N_t)/N$ depends on t alone, then we can solve for $\mu(t)$.

On the other hand, assuming that $\mu = \mu(y)$, we have

$$\frac{\mathrm{d}\mu}{\mathrm{d}y} = \frac{N_t - M_y}{M} \cdot \mu$$

If $(N_t - M_y)/M$ depends on y alone, then we can solve for $\mu(y)$.

If neither case applies, then try some substitutions to transform the ODE into a solvable form.